

AWS ETL Project: Social Media Data Processing using AWS Glue

1. Introduction

This project demonstrates the implementation of an ETL (Extract, Transform, Load) pipeline using cloud services. The goal is to process social media data (tweets and blogs), perform transformations such as joining datasets, extracting hashtags, and generating aggregated insights. The processed data is then stored for analysis.

2. Objective

- To build an ETL pipeline using AWS services
 - To extract data from Amazon S3
 - To transform data using AWS Glue Visual ETL
 - To load processed data into an S3 bucket
 - To analyze data using SQL queries
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3. Tools and Technologies Used

- Amazon S3 (Storage)
 - AWS Glue (ETL Processing)
 - AWS IAM (Access Management)
 - Amazon Athena (Querying)
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4. Architecture Overview

1. Data is uploaded into Amazon S3
 2. AWS Glue Crawlers scan the data and create metadata tables
 3. AWS Glue ETL job processes and transforms the data
 4. Output is stored in another S3 bucket
 5. Data is queried using SQL via Athena
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5. Implementation Steps

5.1 S3 Bucket Creation

- Created bucket: etl-twitter-blog
 - Created folders:
 - etl-social-media (tweets dataset)
 - blog-data (blogs dataset)
 - Uploaded:
 - sample_tweets.csv
 - sample_blogs.csv
 - Created output bucket: etl-cep-output
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5.2 AWS Glue Database

- Created database: social_media_data
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5.3 Classifiers

Two CSV classifiers were created:

- twitter_data
- blog_data

Both configured with:

- Delimiter: comma (,)
 - Quote symbol: "
 - Header: present
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5.4 Crawlers

- Created crawler: tweet-crawl
 - Source: etl-social-media
- Created crawler: blog-crawler
 - Source: blog-data

Both crawlers populated tables:

- sample_tweets_csv

- sample_blogs_csv
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5.5 IAM Role

- Created IAM role: glue-role
 - Attached policy: AdministratorAccess
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5.6 ETL Job (Visual ETL)

Data Sources:

- Tweets dataset
- Blogs dataset

Transformations:

1. Join

- Type: Inner Join
- Condition: user_id

2. Drop Fields

- Removed duplicate user_id

3. Regex Extractor

- Column: tweet_text
- Regex: #(\w+)
- Output: hashtags

4. Aggregate

- Group by: right_user_id
 - Aggregations:
 - Count of tweet_id
 - Minimum of timestamp
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5.7 Output

- Target: S3 bucket etl-cep-output

- Format: JSON
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5.8 Query Execution

Used Amazon Athena to run SQL queries:

Example:

```
SELECT user_id, COUNT(tweet_id)
FROM sample_tweets_csv
GROUP BY user_id;
```

6. Results

- Successfully processed and joined datasets
 - Extracted hashtags from tweets
 - Generated aggregated user-level insights
 - Stored output in S3 for further analysis
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7. Challenges Faced

- Difficulty connecting nodes in Visual ETL
 - Permission issues in S3
 - UI glitches in AWS Glue
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8. Conclusion

This project successfully demonstrates building a scalable ETL pipeline using AWS services. It highlights how cloud-based tools can efficiently process and analyze large datasets with minimal infrastructure setup.

9. Future Enhancements

- Use real-time streaming data
- Add data visualization (QuickSight)
- Automate pipeline scheduling
